

Applied Math Major 2025-2026 Curriculum Chart

Calculus
Complete one sequence

MATH 19A Calculus for Sci., Engr. & Math Prerequisites: Math 3 or Math Placement	&	MATH 19B Calculus for Sci., Engr. & Math Prerequisites: Math 19A or Math 20A
OR		
MATH 20A Honors Calculus Prerequisites: Math Placement Score of 500+ or AP/IB Score Placement	&	MATH 20B Honors Calculus Prerequisites: Math 20A

Linear Algebra & Differential Equations

AM 10 (Strongly Preferred) Math Methods for Engineers I Prerequisites: Math Placement or Math 3	AND	AM 20 (Strongly Preferred) Math Methods for Engineers II Prerequisites: Math 19B or Math 20B and AM 10 or Math 21
OR		
MATH 21 Linear Algebra Prerequisites: Math Placement or Math 3	OR	MATH 24 Differential Equations Prerequisites: Math 22 or Math 23A or AM 30

Discrete Math
Complete One

CSE 16
Applied Discrete Mathematics
Prerequisites: Math 19A or 19B or Math 11B

OR

MATH 100*
Introduction to Proof and Problem Solving
Prerequisites: Math 11A or 19A or 20A, and Math 21 or AM 10

**Note: MATH 100 is preferred for upper division math electives.*

Multivariable Calculus
Complete one sequence

Math 23A Vector Calculus Prerequisites: Math 19A or Math 20A or AP Calc BC Score of 4 or 5	OR	AM 30 (Strongly Preferred) Multivariate Calculus for Engineers Prerequisites: AM 10 or Math 21, and Math 19B or Math 20B
&		
MATH 23B Vector Calculus Prerequisites: Math 23A		

Lower Division Electives

ELECTIVE	ELECTIVE
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A list of the lower division electives can be found on the BE Undergraduate Advising website.

Programming
Complete One

CSE 20
Beginning Programming in Python

OR

CSE 13S
Computer Systems and C Programming

OR

ECE 13
Computer Systems and C Programming

OR

ASTR 19
Practical Programming for the Sciences

OR

ECON 22P
Intro to Python for Data Science

Upper-Division Courses

AM 100 Mathematical Methods for Engineers Prerequisites: AM 20 or Math 24, and AM 30 or Math 23B	AM 112 Introduction to Partial Differential Equations Prerequisite: AM 100 or Permission of Instructor	AM 114 Introduction to Dynamical Systems Prerequisites: AM 10 or MATH 21, AM 20 or MATH 24, and AM 30 or MATH 23A or MATH 22, or PHYS 116A.	AM 129 Foundations of Scientific Computing for Scientists and Engineers Prerequisites: AM 10 or Math 21, and Math 11A or 19A or 20A. Basic programming experience is assumed.	STAT 131 Introduction to Probability Theory Prerequisites: Math 19B or Math 20B. (AM 30 or Math 23B Recommended) OR CSE 107 Probability & Statistics for Engineers Prerequisites: CSE 16 and AM 30 or Math 22 or Math 23A
AM 147 Computational Methods & Applications Prerequisites: AM 10 or Math 21 or Math 148 Computational Methods & Applications Prerequisites: Math 22 or 23A, Math 21 or AM 10, Math 24 or AM 20, and Math 103A or 105A or 152 or AM 147 or CSE 101				

Comprehensive Requirement

AM 170A (DC)
Mathematical Modeling 1
Prerequisites: AM 30, AM 114 or AM 214, Stat 131 or CSE 107, and completion of the Writing and Composition requirements

AND

Choose One

AM 170B
Mathematical Modeling 2
Prerequisites: AM 129 or AM 209, AM 112, AM 147, and AM 170A

OR

AM 195
Senior Thesis Research
Prerequisites: AM 129 or AM 209, AM 112, AM 147, and AM 170A

Upper-Division Electives ♥

ELECTIVE	ELECTIVE	ELECTIVE
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A list of the upper division electives can be found on the BE Undergraduate Advising website.

Applied Math Major 2025-2026 Curriculum Chart

Fall _____	Winter _____	Spring _____	Summer _____

Fall _____	Winter _____	Spring _____	Summer _____

Fall _____	Winter _____	Spring _____	Summer _____

Fall _____	Winter _____	Spring _____	Summer _____

Key Legend

♥ Students are required to take three upper-division elective courses. Up to one of these electives can be replaced by a 5-credit independent study to do research with one of the program faculty. Note that many of these electives have lower-division prerequisites. Students should plan carefully which ones to take to ensure they are prepared for their selected upper-division electives. Also note that enrollment in the graduate courses is by permission of the instructor, who will verify adequate preparation.

Please review the [Applied Math Career Electives](#) document as you select upper division electives that align with your professional goals: <https://docs.google.com/document/d/1dhLayvCKhc4PsO87nT5-apR5h-yIkyfWjYRWSNe2hOE/edit?usp=sharing>

Note:

The prerequisites listed on this curriculum chart are accurate as of July 21, 2025 according to UCSC's general catalog. Prerequisites listed on this chart are subject to change and students should refer to the catalog for the most up to date requirements.